BRAD BURCH

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SUMMARY

Senior full-stack software engineer with 8 years building production systems at Salesforce, Total Brain, Front, and Pawservation. Owns serverless TypeScript systems end to end, from data model and payments through LLM agents, evals, and on-call.

SKILLS

Languages & Frameworks: TypeScript, Python, JavaScript, SQL, Java, Scala, React 19, Node.js, Django, Hono, Spring, Play

Infrastructure: PostgreSQL, SQLite, AWS (DynamoDB), serverless functions, message queues and workflow orchestration (Cloudflare Workers), S3-compatible storage, MSSQL, Docker, Kubernetes, REST APIs, webhooks, distributed systems

AI / LLM: Claude (Haiku, Sonnet), AI agents, Vercel AI SDK, Model Context Protocol (MCP), tool calling, LLM evals, prompt caching, prompt-injection defense, AI cost metering, Claude Code, Cursor

Payments & Security: Stripe (Connect, Billing, Checkout), OAuth 2.1 + PKCE, JWT, SAML 2.0, HMAC webhook verification, AES-GCM, SSRF prevention, multi-tenant isolation

Testing & Delivery: Vitest, Playwright, contract tests, GitHub Actions, Jenkins, CI/CD, observability (structured logging, alerting)

EXPERIENCE

Software Engineer Oct 2022 - Present

Pawservation, San Francisco, CA (pawservation.com) | booking, payments, and AI platform for pet-care businesses

- Built and run the serverless TypeScript platform (React, REST APIs, SQLite) behind a live San Francisco pet-care business: 60+ bookings a month for 30+ active clients, with over half now self-serve through the client portal and an LLM booking agent.
- Replaced 5 to 15 hours a month of manual invoicing, payment reconciliation, and scheduling by integrating Google Calendar, Gmail, Notion, Venmo, and WhatsApp into one booking and payment system.
- Recovered 4 months of failed invoice runs after tracing CPU limit kills to a logo decoded on every PDF; one resize cut CPU time per invoice 88%, from 154 ms to 18 ms.
- Prevented double-booking across 4 concurrent booking channels by serializing every write through a database-backed lock.
- Raised the AI booking agent to 98% of 1,249 eval cases passing every run, and cut eval cost 4.9x by restructuring the prompt so about 95% of input tokens come from cache.
- Rebuilt overnight pricing around a per-night unit, so 92% of overnight stays, like a 22-night sit, bill every night, not one.
- Added monitoring and alerting across integrations and scheduled jobs, so failures get flagged instead of going unnoticed.
- Eliminated the source of 97% of daily database writes by teaching a 15-minute calendar sync to skip unchanged rows.
- Wrote the OAuth 2.1 + PKCE authorization server behind an MCP server, so clients can book from their own AI assistant.
- Closed a prompt-injection hole by rebuilding chat history on the server, so clients can't forge tool results like a $0 quote.

Software Engineer May 2022 - Sep 2022

Front, San Francisco, CA (role eliminated in company-wide layoff)

- Delivered a Node.js REST API for account-level email rules for multi-domain enterprise customers, replacing per-inbox setup.
- Upgraded security in the core JavaScript email library handling 200K+ outbound messages/month.

Software Engineer May 2019 - Sep 2021

Total Brain, San Francisco, CA

- Shortened enterprise onboarding from 2+ weeks to ~3 days by standardizing SAML 2.0 and JWT SSO, HubSpot, and Box integrations into one default path.
- Halved average API response times by migrating MSSQL to DynamoDB with a redesigned schema; mentored junior engineers.
- Scoped integrations live as engineering contact on enterprise Sales and CS calls, running demos and writing onboarding guides.
- Contributed to a 60% lift in new sign-ups by shipping a Scala backend integration for Apple, Google, and Facebook sign-in.

Software Engineer Jul 2017 - Aug 2018

Salesforce, San Francisco, CA

- Led delivery of a config builder and validator used by internal teams, scoping milestones and coordinating partner teams.
- Trimmed internal test setup time 40% with a React management UI, and wrote an OAuth2 security filter for microservice access.

PROJECTS

- **MCP Auth Kit (open source):** OAuth 2.1 + PKCE kit for MCP servers with rate limits, scoped tools, and two-phase confirmation.
- **Roadrunner (open source):** Python/Django and PostgreSQL app matching eBird and iNaturalist sightings to Strava activities via OAuth, webhooks, and a deferred re-check queue.

EDUCATION

DePauw University, Bachelor of Arts in Computer Science 2017